Thawing Frozen Sawlogs Before Debarking: An Energy and Residence-Time Model for the Kuusikkoniemi Hot-Water Pond

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Working paper NIWT-WP 2026/11 · LAUHA winter thaw pond project · 22 September 2026

ABSTRACT. Frozen bark adheres strongly to the stem, so ring debarkers leave bark residue on winter logs and the chips sold to pulp mills fail their bark limit. In the LAUHA winter thaw pond project, Kuusikkoniemi Saha Oy and the Napapiiri Institute of Wood Technology built and tested a two-basin hot-water pond in front of debarking line DB-2 during the winter of 2025–26. This working paper presents four simple equations, calibrated on the trial data, for the thawing energy per cubic metre of log, the residence time in the pond, the daily pond capacity and the heat demand placed on the sawmill's bark-fired boiler. It closes with the operating rules proposed for the winter of 2026–27. The figures are planning values for this site only.

1. Introduction

Kuusikkoniemi Saha Oy operates a sawmill on the west bank of the Kemijoki, 14 km south of Rovaniemi town centre. The mill saws about 214,000 m^3 (solid, under bark) of sawlogs per year, of which 68 % is Scots pine and 32 % Norway spruce, bought mainly from private forests around Ranua, Kemijärvi and Sodankylä. Logs arrive by truck and are stored in the open log yard for 2 to 20 days before sawing. Between November and March their core temperature commonly falls below −20 °C, and in January 2026 the coldest batch logged by the trial reached −34 °C.

The mill has two debarking lines. Line DB-1 handles small logs with a top diameter below 15 cm, which are sold whole as pulpwood or chipped. Line DB-2, a single-rotor ring debarker with a normal feed speed of 70 m/min, prepares all sawlogs for the main saw line. In earlier winters the mill slowed DB-2 to between 35 and 40 m/min whenever the yard temperature was below −15 °C, and still measured an average bark residue of 4.6 % (by area) on debarked pine. The pulp mill that buys the sawmill chips accepts at most 1.0 % bark by dry mass and deducts EUR 3.20 per tonne of chips for each 0.5 percentage point above the limit.

The LAUHA project (budget EUR 1.86 million, of which 40 % came from the Kemijoki Valley Industry Fund and the rest from the company) built a hot-water pond with two parallel concrete basins, A and B, between the log sorting deck and DB-2. The pond was commissioned on 14 October 2025 and the trial ran from 1 December 2025 to 15 March 2026. During that period 3,912 log batches were followed with core temperature probes, and the bark residue after DB-2 was measured on a sample of 25 logs per shift.

2. Thawing energy per cubic metre of log

To debark cleanly, the cambium layer under the bark must be thawed; in practice the mill aims to bring the outer 5 cm of the stem to about +5 °C. The model treats the log as a uniform body and adds three terms: warming the frozen wood to 0 °C, melting the freezable water and warming the thawed wood to the target temperature. Per cubic metre of solid log, the heat required is

$$E = \frac{\rho \times \left[c_f \times \left(0 - T_{log} \right) + \left(u - u_{fs} \right) \times L + c_u \times T_t \right]}{3600} \tag{1}$$

where E is the thawing energy (kWh per m^3 of solid log), ρ the basic density (kg of dry wood per m^3 of green log), c_f and c_u the specific heat of the frozen and unfrozen wet wood (kJ per kg of dry wood per K), T_{log} the core temperature of the log on arrival at the pond (°C), u the moisture content (kg of water per kg of dry wood), u_{fs} the fibre saturation point, taken as 0.30 (water below this level does not freeze), L the latent heat of fusion of ice, 334 kJ/kg, and T_t the target temperature, +5 °C. The divisor 3600 converts kJ to kWh.

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The parameter values measured on Kuusikkoniemi logs in the trial are, for Scots pine: $\rho = 410$ kg/m^3, $u = 0.95$, $c_f = 3.2$ and $c_u = 5.2$; and for Norway spruce: $\rho = 380$ kg/m^3, $u = 0.85$, $c_f = 3.0$ and $c_u = 4.8$. Pine has the higher moisture content because of its wide, wet sapwood, so it needs more energy per cubic metre than spruce at the same temperature.

> **Example 1 (pine at −25 °C).** For a pine batch with $T_{log} = -25$ °C, formula (1) gives $E = 410 \times [3.2 \times 25 + (0.95 - 0.30) \times 334 + 5.2 \times 5] / 3600 = 410 \times (80 + 217.1 + 26) / 3600 = 36.8$ kWh/m^3. Spruce at the same temperature needs 29.8 kWh/m^3.

3. Residence time in the pond

The time a log must stay in the water depends mainly on its diameter and on the water temperature. The trial data were fitted to the empirical form

$$t = \frac{\kappa \times a \times d^2}{T_w} \tag{2}$$

where t is the minimum residence time (hours), κ a cold-log correction factor, equal to 1.00 when the log core temperature is −20 °C or warmer and 1.25 when it is colder than −20 °C, a the species coefficient (h·°C/cm^2), d the mid diameter of the log over bark (cm) and T_w the pond water temperature (°C).

The fitted species coefficients are $a = 0.95$ for Scots pine and $a = 0.80$ for Norway spruce, with a residual standard error of 1.1 hours over 3,912 batches. The first draft of this paper (NIWT-WP 2026/04, April 2026) used $a = 1.10$ for pine, a value taken from laboratory tests on short billets; that value overstated the residence time by about 15 % and is withdrawn. For mixed batches, the operators use the pine coefficient and the largest mid diameter in the batch.

> **Example 2 (pine, coldest week).** A pine batch with a largest mid diameter of 22 cm and a core temperature of −25 °C enters basin A at $T_w = 40$ °C. Because the logs are colder than −20 °C, $\kappa = 1.25$, and $t = 1.25 \times 0.95 \times 22^2 / 40 = 14.4$ hours. The same batch at −15 °C ($\kappa = 1.00$) would need 11.5 hours.

4. Daily pond capacity and heat demand

Each basin holds log baskets with a gross volume of 420 m^3, so the pond has a total gross volume $V_p = 840$ m^3. Logs fill only part of a basket; the trial measured a solid-volume fraction of 0.55. The volume of logs that the pond can thaw per day is

$$V_{day} = \frac{\varphi \times V_p \times 24}{t} \tag{3}$$

where V_{day} is the daily pond capacity (m^3 of solid log per day), φ the solid-volume fraction of a loaded basket (0.55), V_p the gross basket volume of the basins in use (m^3; 840 with both basins, 420 with one) and t the residence time from formula (2) (hours).

The pond water is heated through a plate heat exchanger connected to the sawmill's 6.0 MW bark-fired boiler, which also supplies the drying kilns. The average heating power drawn by the pond is

$$P = \frac{E \times V_{day}}{24 \times \eta} + P_{loss} \tag{4}$$

where P is the average heating power taken from the boiler (kW), E the thawing energy from formula (1) (kWh/m^3), V_{day} the volume of logs thawed per day (m^3/day), η the share of the delivered heat that actually enters the logs (0.78 in the trial; the rest warms the baskets and the water carried out with the logs) and P_{loss} the surface and wall losses of the pond (120 kW with both basins covered; 190 kW if the insulating covers are open).

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Example 3 (both basins, coldest week). With t = 14.4 hours from Example 2 (unrounded 14.37 h), formula (3) gives $V_{day} = 0.55 \times 840 \times 24 / 14.37 \approx 772$ m³ per day. With E = 36.8 kWh/m³ from Example 1, formula (4) gives $P = 36.8 \times 772 / (24 \times 0.78) + 120 \approx 1{,}638$ kW, about 27 % of the boiler rating. Over 24 hours this is 39.3 MWh of heat; at the internal heat transfer price of EUR 24 per MWh used in the LAUHA budget, the pond costs about EUR 943 per day in this week.

The DB-2 saw line consumes about 820 m³ of sawlogs per day on the two-shift winter schedule. In the coldest weeks the pond is therefore slightly short of capacity when only pine is ponded, and the gap is covered from the 1,500 m³ buffer of already debarked logs kept in the covered log hall.

5. Proposed operating rules for winter 2026–27

The following rules are proposed for the pond from 1 November 2026. They were agreed with Kuusikkoniemi Saha Oy on 15 September 2026.

1. **Which logs are ponded.** Pine sawlogs with a core temperature below −4 °C and spruce sawlogs below −8 °C must pass through the pond before DB-2. Spruce sawlogs between −8 °C and −4 °C may go straight to DB-2 if its feed speed is reduced to 45 m/min for the batch. Logs with a top diameter below 15 cm always go to DB-1 and are never ponded, whatever their temperature.
2. **Water temperature.** Pine destined for the export joinery grades may be ponded at no more than 42 °C, because warmer water caused a grey-brown discolouration of the sapwood in the trial; other pine may be ponded at up to 45 °C and spruce at up to 48 °C. During the trial the basins were run at up to 50 °C for all logs until 9 January 2026, when the discolouration was found. Mixed batches follow the lowest limit that applies.
3. **Temperature measurement.** The core temperature of every incoming batch is measured with a probe at 10 cm depth on three logs, and the coldest reading counts. Batches stored for more than 24 hours in the heated log hall are exempt from measurement and are treated as −4 °C.
4. **Single-basin operation.** When one basin is drained for cleaning (planned every 6 weeks, normally on a Sunday), $V_p = 420$ m³ in formula (3), and the pond operator must inform the saw line foreman before the start of the shift so that the debarked-log buffer can be used.

Responsibilities. The pond operator on each shift sets the residence time with formula (2) and records it in the pond log. The water temperature limits in rule 2 are set and changed only by the production manager (Henrik Lehtovaara). Questions on the model go to Dr Aino Rautiainen at the Napapiiri Institute of Wood Technology.

6. Results of the trial and limitations

With the pond in use, the average bark residue on debarked pine fell from 4.6 % to 0.8 % by area, and the bark content of the chips sold to the pulp mill averaged 0.7 % by dry mass, so no price deductions were made after 20 December 2025. DB-2 ran at its normal 70 m/min for 91 % of the trial shifts.

The model has two main limitations. First, formula (1) ignores the heat taken up by snow and ice on the logs, which added up to 2 kWh/m³ after heavy snowfall. Second, the coefficients in formula (2) were fitted to logs with mid diameters of 14 to 38 cm and should not be used outside that range.

References

1. Rautiainen, A. and Kylmälä, J. (2026). *Residence time of frozen sawlogs in a hot-water pond: first estimates*. Working paper NIWT-WP 2026/04, Napapiiri Institute of Wood Technology (withdrawn).
2. Lehtovaara, H. (2025). *LAUHA winter thaw pond: project plan and budget*. Kuusikkoniemi Saha Oy, internal report KS-PR-31.

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